

## SECTION A

### 1 (short)

- (a) An experimentalist passes heat  $Q$  to a mass  $m$  of a perfect gas in a closed rigid vessel. Write down an expression for the change in temperature,  $\Delta T_v$ , of the gas. [1]
- (b) The experimentalist passes the same heat  $Q$  to the same mass  $m$  of the same gas in a cylinder bounded by a frictionless piston that is free to expand against atmospheric pressure,  $p_a$ . Write down an expression for the change in temperature,  $\Delta T_p$ , of the gas. [1]
- (c) If the perfect gas is air, is  $\Delta T_v > \Delta T_p$  or is  $\Delta T_p > \Delta T_v$ ? Why? [2]
- (d) Derive an expression for the work done,  $W$ , by the gas in part (b), in terms of  $Q$  and the ratio of specific heats,  $\gamma$ . Give a physical interpretation of  $(\gamma - 1)/\gamma$ . [4]
- (e) Can you think of a common material (not necessarily a gas) for which  $\Delta T_p$  would be greater than  $\Delta T_v$ ? Explain your answer. [2]